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3 semester , 2 Year

Day 1 Assignment 1

```
import numpy as np

rng = np.random.default_rng()
marks = rng.integers(0, 101, size=(4, 5))

total_marks = np.sum(marks, axis=1)

percentage = (total_marks / (5 * 100)) * 100
print("Marks Matrix:")
print(marks)

print("\nTotal Marks of Each Student:")
print(total_marks)

print("\nPercentage of Each Student:")
print(percentage)
average_marks = np.mean(marks, axis=0)

std_deviation = np.std(marks, axis=0)
print("Marks Matrix (4 Students × 5 Subjects):")
print(marks)

print("\nAverage Marks in Each Subject:")
print(average_marks)
print("\nStandard Deviation in Each Subject:")
print(std_deviation)
```

2. import numpy as np

```
rng = np.random.default_rng()
marks = rng.integers(0, 101, size=(4, 5))

highest_marks = np.max(marks, axis=0)

student_index = np.argmax(marks, axis=0)

print("Marks Matrix (4 Students × 5 Subjects):")
print(marks)

print("\nHighest Marks in Each Subject:")
print(highest_marks)

print("\nIndex of Student Scoring Highest Marks in Each Subject:")
print(student_index)

print("\nStudent Number (1-based indexing):")
print(student_index + 1)
```

```
import numpy as np

rng = np.random.default_rng()
marks = rng.integers(0, 101, size=(4, 5))

pass_mask = marks >= 40

print("Marks Matrix (4 Students × 5 Subjects):")
```

```
print(marks)

print("\nBoolean Mask (True = Pass, False = Fail):")
print(pass_mask)

print("\nPass/Fail Status:")
print(np.where(pass_mask, "Pass", "Fail"))
```